



Multilingual Document OCR Extraction Pipeline

Gls Nexus Hackathon 2026

Batch: 2025-2027

Team ID: 56

Course: MCA Semester 3

Teammate 1: 202500719010044

Teammate 2: 202500719010001

Submission Date: 15-09-2026

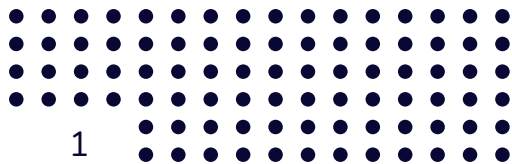
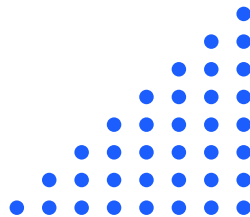


Table of contents

01 Problem Statement

02 Proposed Solution

03 Technology Stack

04 System Workflow

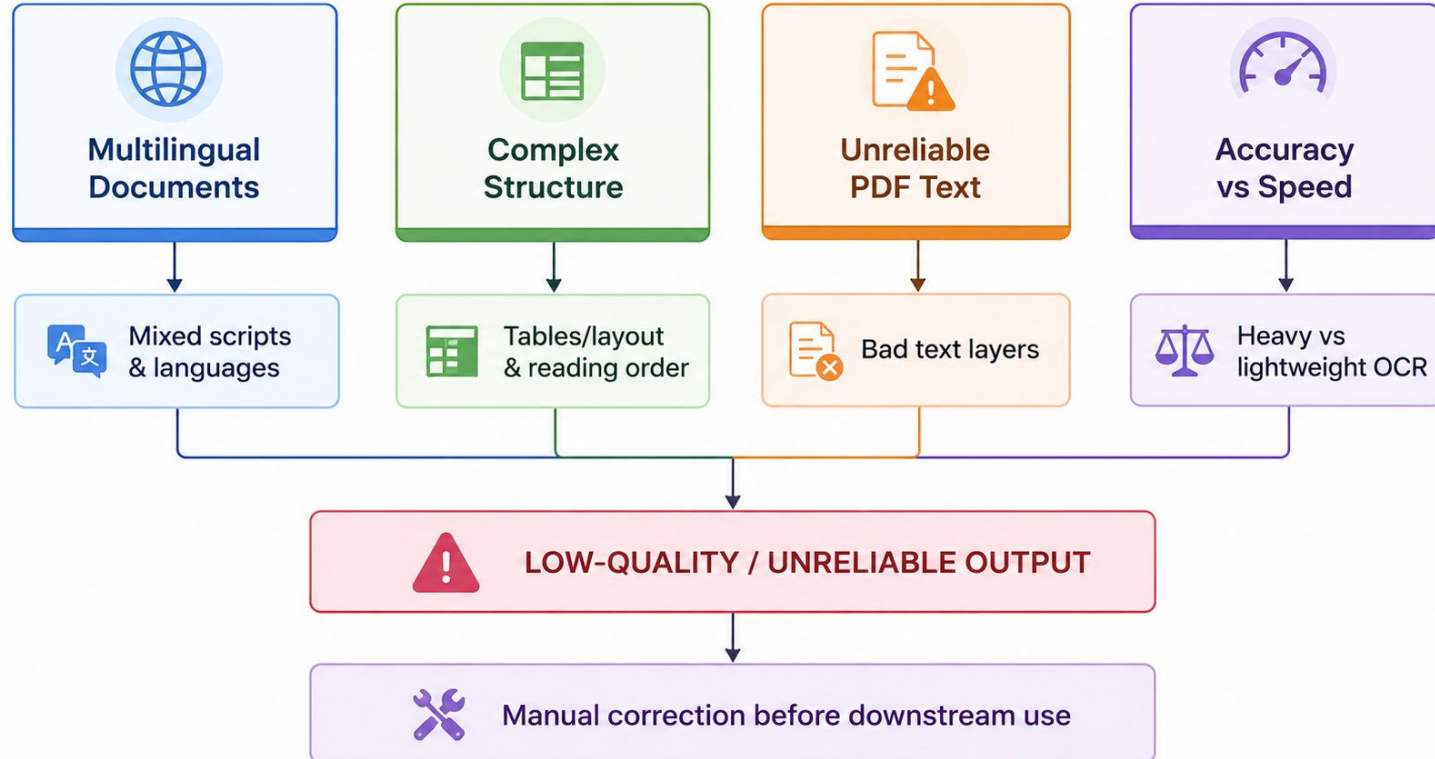
05 Feasibility Analysis

06 Algorithm

07 Assumptions & Challenges

Problem Statement

Title: Multilingual Document OCR Extraction Pipeline



Proposed Solution



1 Document Format Awareness

Detects the uploaded file type and routes it to the appropriate parser (PDF, DOCX, XLSX, PPTX, images, etc.).



2 Multilingual & Script-Aware OCR

Identifies scripts (English, Hindi, Gujarati, etc.) and uses optimized OCR for accurate text extraction.



3 Structure Preservation

Maintains reading order, tables, layouts, and key-value relationships for clean, structured output.



4 Confidence-Based Processing

Accepts reliable results immediately and selectively reprocesses low-confidence regions for better accuracy.



5 Fast Processing Speed

Optimized pipeline for quick processing, with sub-second handling for most standard documents.



6 Easy Integration

Can be easily integrated via REST API or CLI for seamless use in existing systems and workflows.

Proposed Solution

NEXUS INVOICE STATEMENT

Invoice Number: INV-2026-9042

Date: 21/08/2026

Total Amount: \$4,580.00

This document verifies high-throughput multilingual OCR extraction.

```
[NexusOCR] [Document] 'test_digital_english.pdf' | Total Pages: 1 (Processing: 1)
[NexusOCR] -> Page 1/1...
Consider using the pymupdf_layout package for a greatly improved page layout analysis.
[NexusOCR] [Tier 1: Digital PyMuPDF] Score=1.00 | 287.2ms [OK]
[NexusOCR] [Complete] 'test_digital_english.pdf' - 1 pages in 531.8ms

## NEXUS INVOICE STATEMENT

Invoice Number: INV-2026-9042

Date: 21/08/2026

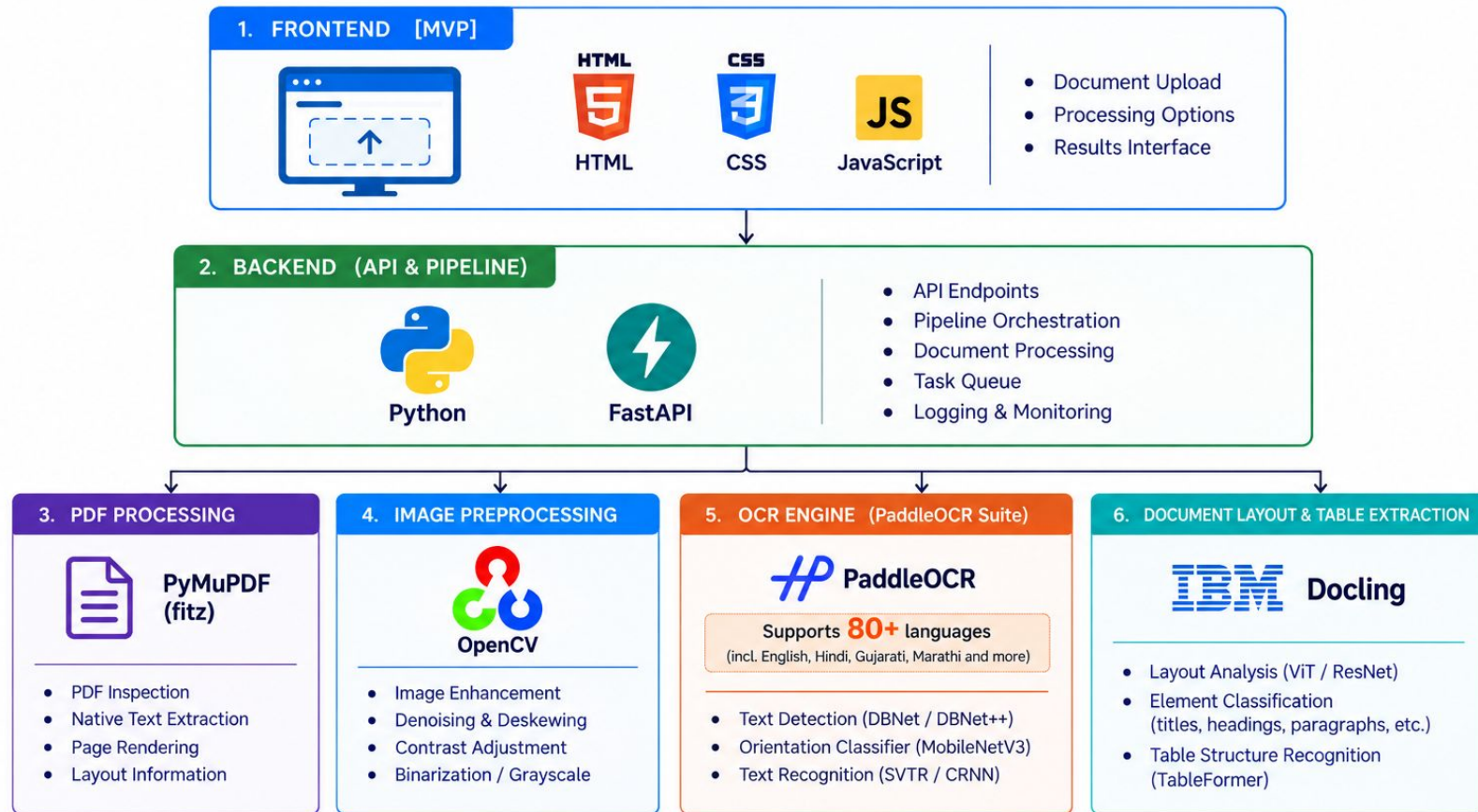
Total Amount: $4,580.00

This document verifies high-throughput multilingual OCR extraction.
(.venv) PS D:\Coding\MCA\nexus-multilingual-ocr-2026> █
```

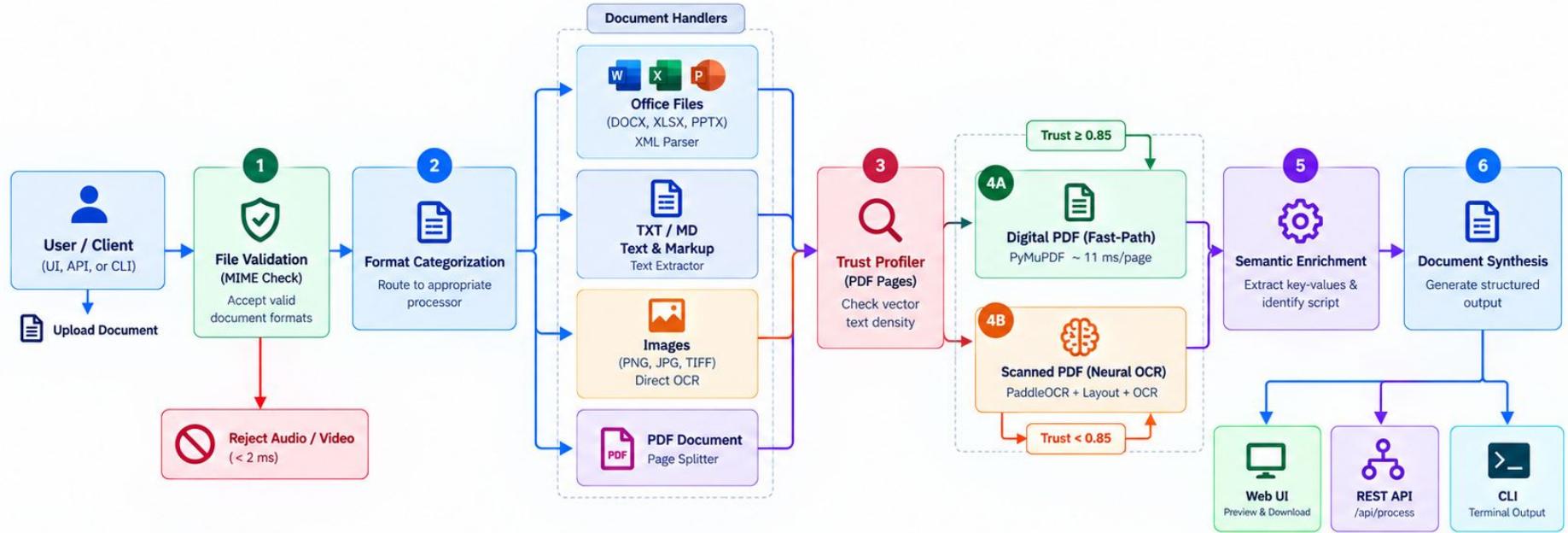
Original Document

Our Solution

Technology Stack



System Workflow



Flow Legend

- Normal Flow
- Conditional Flow
- Rejection Flow
- Merge Flow

Feasibility Analysis

Technical

- Open-source OCR ecosystem.
- Modern OCR architecture.
- Native + Neural OCR Routing.

Market

- High document digitization demand.
- Multilingual OCR gap.
- Multiple industry applications.

Business

- Reduces manual processing.
- Automates downstream workflows.
- Lower dependency on proprietary tools.

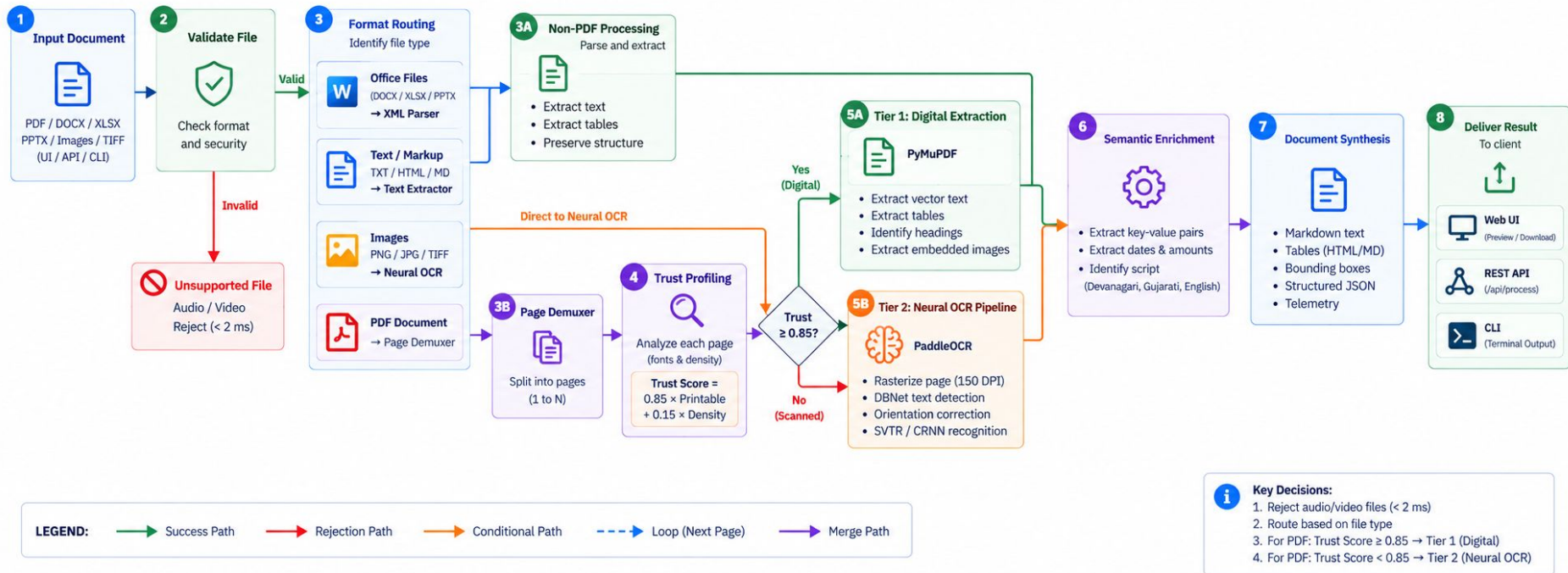
Operational

- Batch processing.
- Configurable pipeline.
- Simple UI + API.

Scalability

- Page-level processing.
- API integration.

Algorithm



Assumptions & Challenges

Assumptions



Documents

Digital, scanned,
hybrid &
degraded PDFs



Languages

English, Hindi
& Gujarati



Text

Printed text is
the focus;
handwriting is
an edge case



Compute

Text documents
with multilingual
support



Efficient

Low cost and
resource-efficient
processing

Assumptions & Challenges

Challenges

01



Poor Scan Quality

Blur, skew, shadows, low contrast and compression can reduce OCR accuracy.

02



Time-Consuming Parsing

Complex office documents (e.g., PPTX) can take longer to parse and extract content compared to simpler formats.

03



Handwriting

Handwritten content is harder to recognize reliably, often requiring additional preprocessing.

04



Accuracy vs Performance

Higher accuracy (e.g., high DPI, complex layouts) increases computational cost, so selective processing is required.

Interface Design



Overview Studio

Adaptive Multilingual Document Pipeline

Precision OCR & Document Intelligence.

Bifurcated architecture delivering sub-10ms digital native text recovery and local neural vision for complex Indic scripts. Built for English, Hindi, and Gujarati with zero external cloud dependencies.

Launch Studio & Extract →

View Benchmarks

15+ Formats

PDF, Scans, Office, Data

2-Tier Hybrid

PyMuPDF + PaddleOCR

Multilingual

Gujarati, Hindi, English

100% Local

Zero Cloud Dependencies

System Architecture & Bifurcation

100% OFFLINE RUNTIME

Multi-Format Document

PAGETRUST PROFILER

Threshold: Score ≥ 0.85

TIER 1: DIGITAL NATIVE

PyMuPDF C-Extension

<10ms · 0MB

TIER 2: NEURAL VISION

PaddleOCR Indic PP-OCR

~100ms · Gujarati/Hindi

UNIFIED DOCUMENT AST

Markdown · Tables · JSON · Images

BENCHMARKS ENGINE TELEMETRY

Speed & Accuracy Telemetry

Empirical benchmarks on local hardware delivering sub-10ms native speed and resilient Indic OCR accuracy.

Extraction Latency Per Page

Empirical execution time per page (lower is better)

10x TO 80x FASTER

Multilingual Accuracy Matrix

High-Res Scans ≥300 DPI · Crisp High-Contrast Print

99.1% AVG

Interface Design

NexusOCR

Overview

Studio

PRESET Hindi & Devanagari Script (Tier 1 Multilingual) ▼

LATENCY 763 ms

TRUST 89%

PAGES 1

test_hindi_devanagari.pdf Selected Document

Change

Re-run

Page 1 / 1

Tier 1: Digital Native (364ms)

BBboxes

100%

Fit

हिंदी दस्तावेज परीक्षण — नेक्सस ओसीआर

दस्तावेज क्रमांक: DL-2026-89432 | दिनांक: 21 अगस्त 2026 | सत्यापन: 100% ट्रस्ट

यह एक डिजिटल हिंदी परीक्षण दस्तावेज है। नेक्सस ओसीआर (NexusOCR) उच्च सटीकता के साथ देवनागरी लिपि, संयुक्त अक्षर, शिरोरेखा और मात्राओं का सटीक विश्लेषण करता है।

तकनीकी विनिर्देश एवं मापदंड (Technical Specifications)

मापदंड (Parameter)	विवरण (Details)	सटीकता (Accuracy)
लिपि पहचान (Script)	देवनागरी (Devanagari / Hindi)	99.2% उच्च परिशुद्धता
प्रसंस्करण टियर (Tier)	टियर 1 डिजिटल फास्ट-पाथ	11 ms (सब-100ms)
हार्डवेयर इंजन (Engine)	AMD CPU स्थानीय निष्पादन	100% ट्रस्ट स्कोर

नेक्सस ओसीआर का मल्टीलिंगुअल इंजन मशरूफ हिंदी और अंग्रेजी दस्तावेजों को बिना किसी तृतीय-पक्ष कलाउड निर्भरता के प्रोसेस करता है।

Markdown

Tables

JSON

Raw Text

Diagnostics

Copy

.md

.json

हिंदी दस्तावेज परीक्षण — नेक्सस ओसीआर

दस्तावेज क्रमांक: DL-2026-89432 | दिनांक: 21 अगस्त 2026 | सत्यापन: 100% ट्रस्ट
यह एक डिजिटल हिंदी परीक्षण दस्तावेज है। नेक्सस ओसीआर (NexusOCR) उच्च सटीकता के साथ देवनागरी लिपि, संयुक्त अक्षर, शिरोरेखा और मात्राओं का सटीक विश्लेषण करता है।
तकनीकी विनिर्देश एवं मापदंड (Technical Specifications):

मापदंड (Parameter)	विवरण (Details)	सटीकता (Accuracy)
लिपि पहचान (Script)	देवनागरी (Devanagari / Hindi)	99.2% उच्च परिशुद्धता
प्रसंस्करण टियर (Tier)	टियर 1 डिजिटल फास्ट-पाथ	11 ms (सब-100ms)
हार्डवेयर इंजन (Engine)	AMD CPU स्थानीय निष्पादन	100% ट्रस्ट स्कोर

नेक्सस ओसीआर का मल्टीलिंगुअल इंजन मिश्रित हिंदी और अंग्रेजी दस्तावेजों को बिना किसी तृतीय-पक्ष कलाउड निर्भरता के प्रोसेस करता है।
नेक्सस ओसीआर 2026 | बहुभाषी दस्तावेज प्रसंस्करण एवं देवनागरी सत्यापन सूट

Extracted 1 page(s) in 763ms

899 chars Scripts: English, Hindi



Thank You

